

FamBench: Protein–Ligand Affinity Models Are Inflated by Protein-Family Redundancy

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Abstract

Co-folding models such as Boltz-2 and Nesso-1 report protein–ligand binding-affinity correlations of Pearson ≈ 0.6 and are increasingly proposed as fast surrogates for free-energy perturbation. We show that this headline accuracy is largely an artifact of *protein-family redundancy* in the test set. Stratifying 3,289 PDBbind complexes by protein-family size (MMseqs2 30% clusters), we find that both models’ accuracy scales monotonically with how heavily the family is represented in the PDB: from Pearson ≈ 0.10 on singleton families to ≈ 0.55 – 0.59 on families with > 300 homologs. A leakage-slope regression (absolute error vs. $\log_{10}$ family size, controlling for the affinity regime) is significantly negative for both co-folders ($t = -7.5$ and -4.3) but *flat* for a random-forest QSAR model and a ligand- k NN baseline evaluated under family-disjoint cross-validation. On novel families the ranking inverts: a shallow, family-disjoint random forest ($r = 0.41$) outperforms both billion-parameter co-folders ($r = 0.32, 0.31$). On a genuinely novel, post-cutoff target (OpenBind EV-A71 2A protease), neither co-folder reliably beats a molecular-weight baseline ($\rho = 0.49$ for Nesso-1, 0.40 for Boltz-2, vs. 0.47 for molecular weight). We release **FamBench**, a plug-in benchmark and Hugging Face dataset that measures affinity accuracy *as a function of protein-family novelty* rather than as a single correlation, so that models are judged on the case that matters for a new drug program: a target they have not seen.

1 Introduction

Predicting protein–ligand binding affinity from structure is a central problem in computational drug discovery. A recent wave of *co-folding* models—AlphaFold3 [1], Boltz-1/2 [2, 3], Chai-1 [4], and coarse-grained successors such as Nesso-1 [5] and TerraBind [6]—predict the bound complex and, increasingly, the affinity directly. Boltz-2 reports correlations approaching free-energy perturbation (FEP) at a fraction of the cost [3], and these numbers are driving adoption in hit identification and lead optimization.

Such models are almost always summarized by a *single* correlation between predicted and measured affinity on a held-out set. But a single number cannot distinguish a model that has learned the physics of binding from one that recognizes protein families it saw many times in the Protein Data Bank (PDB). The distinction is not academic: a medicinal chemistry program succeeds or fails on *novel* targets, precisely those least represented in public structural data.

A large literature has documented that ligand-based and docking benchmarks reward memorization [7, 8], that affinity networks trained on PDBbind rely on train–test similarity rather than interaction physics [9, 10], and that re-splitting the data to remove leakage lowers apparent accuracy

[11]. What is missing is a *quantitative, model-agnostic account* of how co-folding affinity accuracy depends on the one axis that matters most for prospective use—*protein-family novelty*—and a benchmark that measures it directly.

We make three contributions.

1. **We show co-folding affinity accuracy is a redundancy artifact.** On a matched set of 3,289 PDBbind complexes, Boltz-2 and Nesso-1 accuracy rises monotonically with protein-family size, collapsing toward zero on singleton families (Sec. 5.1, Fig. 1).
2. **We isolate the mechanism with controls.** A random-forest QSAR model and a ligand- k NN baseline, evaluated under *family-disjoint* cross-validation, show no such gradient; on novel families they *outperform* the co-folders (Sec. 5.2). On a genuinely novel post-cutoff target, neither co-folder beats molecular weight (Sec. 5.3).
3. **We release FamBench**, a plug-in Python/CLI benchmark and Hugging Face dataset with confound-guarded metrics, so any affinity model can be scored on its family-novelty curve in a few lines (Sec. 3).

2 Related Work

Co-folding affinity models. The AlphaFold2 breakthrough [12] was extended to protein–ligand complexes by AlphaFold3 [1] and open reimplementations such as Boltz-1 [2], Chai-1 [4], and RoseTTAFold-All-Atom [13]. A second generation targets affinity directly: Boltz-2 [3] is, to our knowledge, the first co-folder to approach FEP accuracy, and Nesso-1 [5] and TerraBind [6] follow with faster coarse-grained representations. They are evaluated by overall correlation, inviting the generalization questions that have long dogged scoring functions.

Scoring and QSAR baselines. Empirical (AutoDock Vina [14], smina [15]) and machine-learned (RF-Score [16], Gnina [17], AEV-PLIG [18]) scoring functions established learning affinity from structure. Nearly all are trained and tested on PDBbind [19] and CASF-2016 [20], with labels from BindingDB [21] and ChEMBL [22], against the FEP+ reference [23].

Leakage and benchmark critiques. Wallach and Heifets [7] showed most ligand-based benchmarks reward memorization, and Chen et al. [8] traced deep-learning “success” on DUD-E to dataset bias. For affinity, Volkov et al. [9] found ligand-only and protein-only models match full-complex models, and nearest-training-neighbor baselines are already strong—evidence of memorization. Kanakala et al. [10] confirmed latent biases in popular sets. The community response has been to re-split: Leak-Proof PDBBind [11] removes cross-contamination and shows retrained models generalize better; curated resources such as MISATO [24] follow suit.

Time-split and out-of-distribution evaluation. Sheridan [25] argued that time-split, not random split, estimates prospective accuracy. The most stringent test is a target released after a model’s training cutoff; the OpenBind EV-A71 2A protease release [26] provides exactly such a post-2021 target, free of PDBbind contamination.

Our contribution. Prior work establishes that benchmarks reward memorization and that re-splitting mitigates it. Unlike LP-PDBBind [11], which fixes a split and reports one aggregate number, we treat *redundancy as the independent variable* and expose the accuracy-versus-novelty curve directly, for the newest class of models (co-folders) that no prior leakage study covers.

3 The FamBench Benchmark

FamBench comprises two evaluation sets and a set of confound-guarded metrics.

Redundancy set. 18,759 protein–ligand complexes derived from PDBbind [19] with an exact (=) K_d , K_i , or IC_{50} label converted to $\text{p}K = -\log_{10}(\text{conc})$. For each complex we record the protein sequence, ligand SMILES, and—critically—the *protein-family size*: the number of complexes in the same MMseqs2 [27] cluster at 30% identity / 80% coverage. Family sizes range from 1 (890 singletons) to 809, over 2,203 families. We define six redundancy bins (1, 2–5, 6–20, 21–80, 81–300, 301+) and a balanced “core” subset of 3,360 complexes (560 per bin) for quick evaluation. A ligand nearest-neighbor Tanimoto (ECFP4) is also recorded as a secondary novelty axis.

Temporal set. 649 compounds with measured K_d against a single, structurally novel, post-2021 target: the Enterovirus A71 2A protease from the OpenBind consortium release [26] (CC0). Because the target post-dates the 2021-09-30 training cutoff of the models we evaluate, it is a genuine out-of-distribution test. We ship the consortium’s molecular-weight, clogp, Gnina, smina, AEV-PLIG, and AQ-Affinity predictions as baselines.

Metrics. The primary quantity is not a single correlation but its dependence on family size. For each redundancy bin we report Pearson r between predicted and measured affinity within a *matched* $\text{p}K$ window [4.5, 8.0], which holds the label spread (and therefore the maximum attainable correlation) approximately constant across bins. We summarize the whole curve with two headline numbers:

- **Novel-family Pearson:** accuracy on families of size ≤ 5 (the number a model should report for a new target).
- **Leakage slope:** the coefficient of $\log_{10}(\text{family size})$ in an ordinary least-squares regression of absolute error on family size, the affinity regime ($\text{p}K$, $\text{p}K^2$), and affinity type. A significantly negative slope means accuracy is *bought by redundancy*.

Per-bin confidence intervals are non-parametric bootstrap (2,000 resamples). On the temporal set we report within-target Spearman ρ and the residual ρ after linearly removing molecular weight, which quantifies signal *beyond* ligand size.

4 Experimental Setup

We evaluate six methods spanning three families. **Co-folders:** Boltz-2 [3] (with cuEquivariance kernels and ColabFold MSAs) and Nesso-1 [5] (ESM-2 embeddings [28]), both run from public weights with their default settings; both output $\log_{10}(\text{IC}_{50})$, which we negate to the FamBench “higher-is-stronger” convention. **Shallow learned baselines:** a random-forest QSAR model (RF-QSAR) on ECFP4 (1024 bit) plus amino-acid composition, and a ligand- k NN memorization baseline ($k=5$, ECFP4 Tanimoto). Both are evaluated under *5-fold family-disjoint* (GroupKFold on family id) cross-validation, so their predictions are always on held-out families—the honest counterpart to the co-folders, whose test families *are* present in their PDB training data. **Trivial baselines:** molecular weight and clogp. All methods are scored on the same 3,289-complex intersection (Boltz-2 on a 578-complex subset owing to its higher per-target cost).

Table 1: Affinity accuracy on the 3,289-complex matched set. Overall and per-stratum Pearson r ; the *leakage slope* is the coefficient (with t -statistic) of $\log_{10}(\text{family size})$ in the absolute-error regression. Co-folders have a large, significant negative slope; family-disjoint and trivial baselines do not. Boltz-2 is scored on a 578-complex subset.

Method	Type	Overall r	Novel-fam. r	Redund.-fam. r	Leakage slope (t)
Nesso-1	co-folder	0.588	0.324	0.599	-0.109 (-7.5)
Boltz-2	co-folder	0.467	0.308	0.429	-0.139 (-4.3)
RF-QSAR	family-disjoint ML	0.479	0.411	0.320	$+0.008$ ($+0.6$)
ligand- k NN	family-disjoint ML	0.407	0.350	0.374	-0.002 (-0.1)
clogp	trivial	0.284	0.121	0.234	$+0.010$ ($+1.0$)
mol. weight	trivial	0.069	0.119	0.184	$+0.017$ ($+2.0$)

Table 2: Matched-p K Pearson r by protein-family size. The co-folders rise steeply with redundancy; the family-disjoint baselines are flat. ($n \approx 350$ per Nesso-1 bin.)

Method	protein-family size					
	1	2-5	6-20	21-80	81-300	301+
Nesso-1	0.105	0.225	0.378	0.421	0.437	0.551
Boltz-2	-0.030	-0.007	-0.008	0.420	0.371	0.585
RF-QSAR	0.259	0.264	0.257	0.254	0.196	0.177
ligand- k NN	0.215	0.223	0.192	0.236	0.174	0.180
mol. weight	-0.032	0.009	-0.101	0.107	0.070	-0.010

5 Results

5.1 Co-folding accuracy scales with protein-family redundancy

Table 1 reports overall, novel-family, and redundant-family accuracy, together with the leakage slope. Overall, the co-folders lead (Nesso-1 $r = 0.588$), consistent with their published numbers. But that lead is concentrated on redundant families: Nesso-1 reaches $r = 0.599$ on families with ≥ 81 homologs and only $r = 0.324$ on families of size ≤ 5 . Boltz-2 shows the same pattern more sharply (0.429 vs. 0.308).

The matched-p K dose-response (Table 2, Fig. 1) makes the effect unambiguous. With label spread held constant across bins, Nesso-1 rises from $r = 0.105$ on singletons to 0.551 on the largest families; Boltz-2 is statistically *zero* on all families of size ≤ 20 ($r = -0.03$ to -0.01) and only acquires signal above 21 homologs. Because the window pins the attainable correlation, this gradient cannot be explained by narrower affinity ranges in small families.

5.2 Controls isolate leakage as the mechanism

The two shallow baselines are the diagnostic. Evaluated under family-disjoint cross-validation—so they can never have “seen” the test family—RF-QSAR and ligand- k NN have leakage slopes indistinguishable from zero ($t = +0.6$ and -0.1 ; Fig. 2) and flat dose-response curves (Table 2). The gradient is therefore not an intrinsic property of “harder” novel targets (which would penalize every method equally); it appears *only* for models whose training data contains the test families. When family leakage is removed by construction, the gradient vanishes.

The practical consequence is a rank inversion (Fig. 2). On novel families (size ≤ 5), the family-disjoint random forest ($r = 0.411$) *outperforms* both co-folders (Nesso-1 0.324, Boltz-2 0.308), and

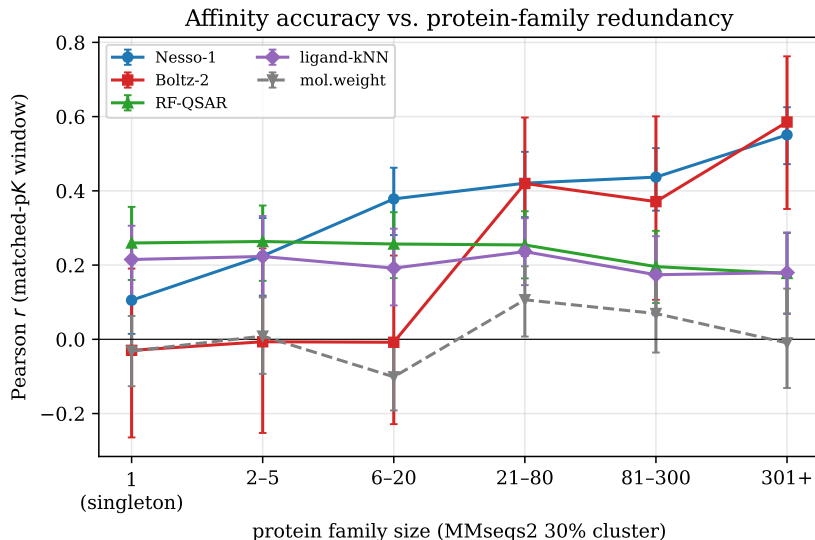


Figure 1: Affinity accuracy versus protein-family redundancy (matched-pK window, bootstrap 95% CIs). Co-folders (Nesso-1, Boltz-2) rise steeply with family size; a family-disjoint random forest and ligand- k NN are flat. The co-folders’ apparent superiority is confined to redundant families.

Table 3: Novel-target ranking (OpenBind EV-A71 2A protease, 643 compounds). Spearman ρ with experiment, and ρ after removing molecular weight. No method reliably beats the molecular-weight baseline.

Method	Spearman ρ	ρ beyond mol. weight
Nesso-1	0.486	0.233
molecular weight	0.469	—
Gnina	0.431	−0.082
Boltz-2	0.395	0.112
smina	0.229	−0.009
AEV-PLIG	0.228	−0.143
clogp	0.163	−0.057
AQ-Affinity	0.134	−0.019

even ligand- k NN (0.350) is competitive. A model with 10^3 parameters and no structure beats billion-parameter co-folders on exactly the regime that matters for a new target.

5.3 On a novel target, co-folders do not beat molecular weight

The temporal set removes any ambiguity about “family size” as a proxy: the EV-A71 2A protease post-dates the models’ training cutoff and shares no family with PDBbind at the evaluation threshold. Table 3 reports within-target Spearman ρ over 643 compounds. Molecular weight alone achieves $\rho = 0.469$. Nesso-1 barely exceeds it (0.486); Boltz-2 *underperforms* it (0.395), as do Gnina, smina, AEV-PLIG, AQ-Affinity, and clogp. After residualizing on molecular weight, Nesso-1 retains a modest $\rho = 0.233$ of genuine signal and Boltz-2 only 0.112; the docking and other ML scorers retain essentially none. On a target no model has seen, sophisticated affinity prediction collapses to ligand size.

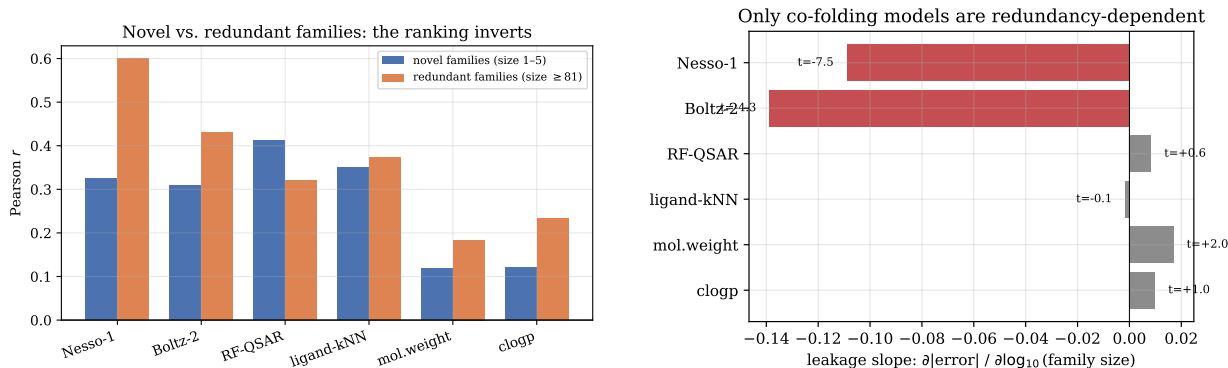


Figure 2: **Left:** novel- vs. redundant-family Pearson r . The co-folders’ advantage is entirely on redundant families; on novel families the ranking inverts. **Right:** leakage slope (change in $|\text{error}|$ per decade of family size). Only the co-folders are significantly redundancy-dependent ($|t| > 2$, red).

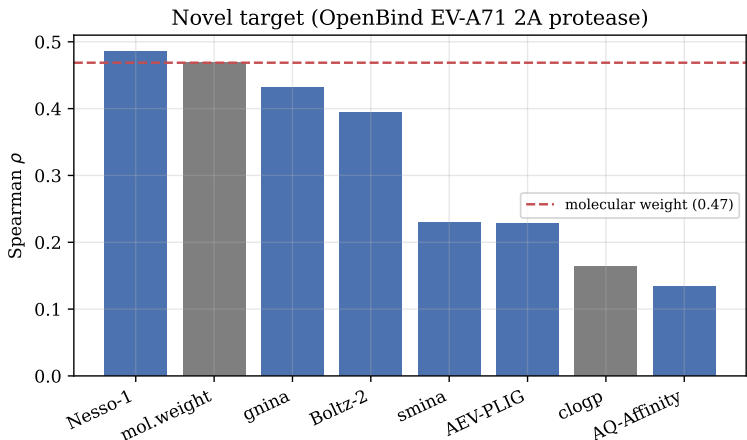


Figure 3: Within-target Spearman ρ on the novel OpenBind target. The dashed line is the molecular-weight baseline; only Nesso-1 marginally exceeds it.

6 Discussion

Three findings cohere into a single message. First, co-folding affinity accuracy is a *monotone function of protein-family redundancy*, not a fixed property of the model. Second, the effect is specifically *leakage*: matched controls that cannot memorize the family show no gradient, and on novel families they win. Third, the endpoint of the gradient—a genuinely novel target—is a regime where the state of the art does not beat molecular weight.

This does not imply co-folders are useless; on well-populated families they are the most accurate methods we tested. It implies that the standard practice of reporting a single overall correlation *systematically overstates prospective accuracy*, because public test sets are dominated by redundant families. A reader who sees “Pearson 0.6” should ask “on which families?”—the honest number for a new target is closer to 0.1–0.3. Our controls also suggest a cheap, strong prospective baseline that is rarely reported: a family-disjoint QSAR model.

Limitations. The redundancy set is derived from PDBbind and is therefore pre-cutoff; its leakage is redundancy-based, and we use family size as a proxy for training representation. The temporal set removes this proxy but is a single target with a congeneric compound series, so its absolute numbers should not be over-generalized. We evaluate public weights at default settings and do not retrain the co-folders on de-leaked data; whether retraining closes the gap (as for classical scorers [11]) is open. Finally, family size at 30% identity is one redundancy axis; ligand-scaffold redundancy is a complementary axis we record but do not analyze here.

7 Reproducibility and Artifact

FamBench is released as an installable package (`pip install fambench`) with a Hugging Face dataset. Scoring a model requires wrapping it as `predict(sequence, smiles) → float` and calling `run_redundancy` and `run_temporal`; a language-agnostic CLI (`fambench score`) scores a predictions CSV. All metrics, confound guards (matched- pK windows, leakage-slope regression, bootstrap CIs), baselines, and the dataset-construction pipeline are included. Every number in this paper is reproduced by the shipped harness from the released predictions.

8 Conclusion

Binding-affinity leaderboards for co-folding models measure the wrong thing: accuracy on protein families the model has already seen. FamBench re-centers evaluation on protein-family novelty, exposes a steep redundancy gradient in Boltz-2 and Nesso-1, and shows that on a novel target neither beats molecular weight. We hope reporting the novel-family number and the leakage slope—rather than a single correlation—becomes standard practice for the next generation of affinity models.

Data and code availability. Benchmark, code, and figures: <https://github.com/mrnaFold/fambench>. Dataset: <https://huggingface.co/datasets/mrnaFold/fambench>.

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